

Homework 10

Problem 1. Consider $\mathbf{G}(n, p)$ with $p = \frac{1}{3n}$.

Use the second moment method to show that with high probability there exists a simple path of length 10.

Solution. [Use the second moment method]

$$I_\ell = \begin{cases} 1 & \ell \text{ is a length 10 simple path} \\ 0 & \text{otherwise} \end{cases}$$

X is the number of simple path, then $X = \sum_\ell I_\ell$.

The expectation would be $E(X) = \frac{1}{2}(n)_{11} \times p^{10}$. Thus

$$(E(X))^2 = \frac{1}{4} [(n)_{11}]^2 \times p^{20} \quad (\star)$$

(Note $(\star)$ is about $\Theta(n^2)$.)

Then to calculate $E(X^2)$.

$$E(X^2) = E\left[\left(\sum_\ell I_\ell\right)^2\right] = E\left[\sum_\ell I_\ell \sum_{\ell'} I_{\ell'}\right] = \sum_{\ell, \ell'} E(I_\ell I_{\ell'}) \quad (\Delta)$$

ℓ and ℓ' will be independent to each other unless they have common vertices or edges. We use $k = |\ell \cap \ell'|$ to stand for the number of common edges between ℓ and ℓ' , and $s = |\ell \cap \ell'|$ to stand for the number of common vertices used by ℓ and ℓ' . Obviously $(0 \leq k \leq 10) \wedge (0 \leq s \leq 11) \wedge (k \geq 1 \rightarrow s \geq k + 1)$.

(Δ) can be divided into the following subcases:

1. $k = 0$

$$(a) \quad s = 0: \sum_{\ell, \ell'} E(I_\ell I_{\ell'}) = \frac{1}{8}(n)_{22} \times p^{20} \leq (E(X))^2;$$

(b) $1 \leq s \leq 11$:

for each s , $\sum_{\ell, \ell'} E(I_\ell I_{\ell'}) = c \cdot (n)_{22-s} \times (p)^{20} = \mathbf{o}((E(X))^2)$, where c is a constant number.

2. $1 \leq k \leq 10$ ($2 \leq s \leq 11$)

The general formula of each of these cases (constant many) would be

$$\begin{aligned}
 \sum_{\ell, \ell'} E(I_{\ell} I_{\ell'}) &= d \cdot (n)_{22-s} \times p^{20-k} \\
 &\leq d \cdot (n)_{22-s} \times p^{20-(s-1)} \\
 &= d \cdot (n)_{22-s} \times p^{21-s} \\
 &= \mathbf{o}((E(X))^2)
 \end{aligned}$$

Combining the above results we get that $Var(X) = E(X^2) - (E(X))^2 = \mathbf{o}((E(X))^2)$.
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